

Chapter 2: Interpolation and Approximation:

① Lagrange's Interpolation Technique: (P.C) Python

1. Using Lagrange's Interpolation technique, estimate y for $x=3$ from the given data.

$$x: 2 \quad 4 \quad 7 \quad 9$$

$$y: 1 \quad 2 \quad 1 \quad 3$$

sol: Here,

$$x: 2 \quad 4 \quad 7 \quad 9$$

$$y: 1 \quad 2 \quad 1 \quad 3$$

By Lagrange's interpolation method,

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)y_0}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} + \frac{(x-x_0)(x-x_2)(x-x_3)y_1}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} \\ + \frac{(x-x_0)(x-x_1)(x-x_3)y_2}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} + \frac{(x-x_0)(x-x_1)(x-x_2)y_3}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}$$

$$\Rightarrow y(x) = \frac{(x-4)(x-7)(x-9)}{(2-4)(2-7)(2-9)} \times 1 + \frac{(x-2)(x-7)(x-9)}{(4-2)(4-7)(4-9)} \times 2$$

$$+ \frac{(x-2)(x-4)(x-9)}{(7-2)(7-4)(7-9)} \times 1 + \frac{(x-2)(x-4)(x-7)}{(9-2)(9-4)(9-7)} \times 3$$

$$\Rightarrow y(x) = -\frac{1}{70} \left[(x-4)(x^2-16x+63) \right] + \frac{1}{15} \left[(x-2)(x^2-16x+63) \right] \\ - \frac{1}{30} \left[(x-2)(x^2-13x+36) \right] + \frac{3}{70} \left[(x-2)(x^2-11x+28) \right]$$

$$\Rightarrow y(x) = -\frac{1}{70} \left[x^3 - 16x^2 + 63x - 4x^2 + 64x - 252 \right] \\ + \frac{1}{15} \left[x^3 - 16x^2 + 63x - 2x^2 + 32x - 126 \right] \\ - \frac{1}{30} \left[x^3 - 13x^2 + 36x - 2x^2 + 26x - 72 \right] \\ + \frac{3}{70} \left[x^3 - 11x^2 + 28x - 2x^2 + 22x - 56 \right]$$

$$\Rightarrow y(x) = -\frac{1}{70} [x^3 - 20x^2 + 127x - 252] + \frac{1}{15} [x^3 - 18x^2 + 95x - 126] \\ - \frac{1}{30} [x^3 - 15x^2 + 62x - 72] + \frac{3}{70} [x^3 - 13x^2 + 50x - 56]$$

$$\Rightarrow y(x) = \left(-\frac{1}{70} + \frac{1}{15} - \frac{1}{30} + \frac{3}{70}\right)x^3 + \left(\frac{20}{70} - \frac{18}{15} + \frac{15}{30} - \frac{39}{70}\right)x^2 \\ + \left(-\frac{127}{70} + \frac{95}{15} - \frac{62}{30} + \frac{150}{70}\right)x + \left(\frac{252}{70} - \frac{126}{15} + \frac{72}{30} - \frac{168}{70}\right)$$

$$\Rightarrow y(x) = \frac{13}{210}x^3 - \frac{34}{35}x^2 + \frac{193}{42}x - \frac{24}{5}$$

This is the required interpolating polynomial. Ans

At $x=3$:

$$y(3) = \frac{13}{210} \times 3^3 - \frac{34}{35} \times 3^2 + \frac{193}{42} \times 3 - \frac{24}{5}$$

$$\therefore y(3) = \frac{67}{35} \approx 1.9143 \quad \underline{\underline{\text{Ans}}}$$

② Using Lagrange's interpolation technique, find an interpolating polynomial from the given data:

$$x: 0 \quad 1 \quad 3 \quad 4$$

$$y: 21 \quad 15 \quad 12 \quad 10$$

Estimate $y(2)$.

sol Here,

$$x: 0 \quad 1 \quad 3 \quad 4$$

$$y: 21 \quad 15 \quad 12 \quad 10$$

By Lagrange's interpolation formula,

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} \times y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} \times y_1$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} \times y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} \times y_3$$

$$\Rightarrow y(x) = \frac{(x-1)(x-3)(x-4)}{(0-1)(0-3)(0-4)} x^{21} + \frac{(x-0)(x-3)(x-4)}{(1-0)(1-3)(1-4)} x^{15} \\ + \frac{(x-0)(x-1)(x-4)}{(3-0)(3-1)(3-4)} x^{12} + \frac{(x-0)(x-1)(x-3)}{(4-0)(4-1)(4-3)} x^{10}$$

$$\Rightarrow y(x) = -\frac{7}{4} [(x-1)(x^2-7x+12)] + \frac{5}{2} [x(x^2-7x+12)] \\ - 2 [x(x^2-5x+4)] + \frac{5}{6} [x(x^2-4x+3)]$$

$$\Rightarrow y(x) = -\frac{7}{4} [x^3-7x^2+12x-x^2+7x-12] + \frac{5}{2} [x^3-7x^2+12x] \\ - 2 [x^3-5x^2+4x] + \frac{5}{6} [x^3-4x^2+3x]$$

~~$$\Rightarrow y(x) = \left(-\frac{7}{4} + \frac{5}{2} - 2 + \frac{5}{6}\right) x^3 + \left(\frac{49}{4} - \frac{35}{2} + 10 - \frac{20}{6}\right) x^2$$~~

$$\Rightarrow y(x) = -\frac{7}{4} (x^3-8x^2+19x-12) + \frac{5}{2} (x^3-7x^2+12x) \\ - 2 (x^3-5x^2+4x) + \frac{5}{6} (x^3-4x^2+3x)$$

$$\Rightarrow y(x) = \left(-\frac{7}{4} + \frac{5}{2} - 2 + \frac{5}{6}\right) x^3 + \left(\frac{56}{4} - \frac{35}{2} + 10 - \frac{20}{6}\right) x^2 \\ + \left(\frac{133}{4} + \frac{60}{2} - 8x + \frac{15}{6}\right) x + \left(\frac{84}{4}\right)$$

$$\Rightarrow y(x) = -\frac{5}{12} x^3 + \frac{19}{6} x^2 - \frac{35}{4} x + 21$$

$$\therefore y(x) = -\frac{5}{12} x^3 + \frac{19}{6} x^2 - \frac{35}{4} x + 21$$

✓ This is the required interpolating polynomial.

At $x=2$:

$$y(2) = -\frac{5}{12} \times 2^3 + \frac{19}{6} \times 2^2 - \frac{35}{4} \times 2 + 21$$

$$\therefore y(2) = \frac{77}{6} \quad \underline{\underline{Ans}}$$

Q.07 A curve passes through the points $(0, 18)$, $(1, 10)$, $(3, -2)$ and $(6, -12)$. Using a suitable interpolation formula estimate y for $x = 4$.

Here, $x: 0 \quad 1 \quad 3 \quad 6$
 $y: 18 \quad 10 \quad -2 \quad -12$

By Lagrange's interpolation technique:

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} x y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} x y_1$$
$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} x y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} x y_3$$

$$\Rightarrow y(x) = \frac{(x-1)(x-3)(x-6)}{(0-1)(0-3)(0-6)} x 18 + \frac{(x-0)(x-3)(x-6)}{(1-0)(1-3)(1-6)} x 10$$
$$+ \frac{(x-0)(x-1)(x-6)}{(3-0)(3-1)(3-6)} x -2 + \frac{(x-0)(x-1)(x-3)}{(6-0)(6-1)(6-3)} x -12$$

$$\Rightarrow y(x) = -1 \left[(x-1)(x^2 - 9x + 18) \right] + 1 \left[x(x^2 - 9x + 18) \right]$$
$$+ \frac{1}{9} \left[x(x^2 - 7x + 6) \right] - \frac{2}{15} \left[x(x^2 - 4x + 3) \right]$$

$$\Rightarrow y(x) = -1 \left[x^3 - 9x^2 + 18x - x^2 + 9x - 18 \right] + 1 \left[x^3 - 9x^2 + 18x \right]$$
$$+ \frac{1}{9} \left[x^3 - 7x^2 + 6x \right] - \frac{2}{15} \left[x^3 - 4x^2 + 3x \right]$$

$$\Rightarrow y(x) = -1 \left[x^3 - 10x^2 + 27x - 18 \right] + \left[x^3 - 9x^2 + 18x \right]$$
$$+ \frac{1}{9} \left[x^3 - 7x^2 + 6x \right] - \frac{2}{15} \left[x^3 - 4x^2 + 3x \right]$$

$$\Rightarrow y(x) = \left[-1 + 1 + \frac{1}{9} - \frac{2}{15} \right] x^3 + \left[10 - 9 - \frac{7}{9} + \frac{8}{15} \right] x^2$$
$$+ \left[-27 + 18 + \frac{6}{9} - \frac{6}{15} \right] x + 18$$

$$\Rightarrow y(x) = -\frac{1}{45}x^3 + \frac{34}{45}x^2 - \frac{131}{15}x + 18$$

$$\therefore y(x) = -\frac{1}{45}x^3 + \frac{34}{45}x^2 - \frac{131}{15}x + 18$$

This is the required interpolating polynomial. Ans

At $x = 4$:

$$y(4) = -\frac{1}{45} \times 4^3 + \frac{34}{45} \times 4^2 - \frac{131}{15} \times 4 + 18$$

$$\therefore y(4) = -\frac{886}{315} \approx -2.8127 \quad \text{Ans}$$

$$\Rightarrow y(4) = -\frac{94}{15}$$

$$\therefore y(4) = -6.2667 \quad \text{Ans}$$

② Newton's Divided Difference Formula:

1) Using Newton's divided difference formula, find the interpolating polynomial and hence estimate $y(10)$ from the following data:

$$x : 5 \quad 6 \quad 9 \quad 11$$

$$y : 12 \quad 13 \quad 14 \quad 16$$

Here,

$$x : 5 \quad 6 \quad 9 \quad 11$$

$$y : 12 \quad 13 \quad 14 \quad 16$$

The Newton's divided difference formula is given by

$$y(x) = y_0 + (x-x_0) y[x_0, x_1] + (x-x_0)(x-x_1) y[x_0, x_1, x_2] + (x-x_0)(x-x_1)(x-x_2) y[x_0, x_1, x_2, x_3] \quad \text{--- (1)}$$

Divided Difference Table:

x	y	I DD	II DD	III DD
5	12	1		
6	13		-0.1667	
9	14	0.3333	0.1333	0.0500
11	16	1		

From (1),

$$y(x) = 12 + (x-5) \cdot 1 + (x-5)(x-6) \cdot (-0.1667) + (x-5)(x-6)(x-9) \cdot 0.0500$$

$$\Rightarrow y(x) = 12 + x - 5 + (x^2 - 11x + 30) \cdot (-0.1667) + (x^3 - 15x^2 + 54x - 5x^2 + 75x - 270) \cdot 0.0500$$

$$\Rightarrow y(x) = 7 + x + (x^2 - 11x + 30) \cdot (-0.1667) + (x^3 - 15x^2 + 54x - 5x^2 + 75x - 270) \cdot 0.0500$$

$$\Rightarrow y(x) = 7 + x - 0.1667x^2 + 1.8337x - 5.0010 + (x^3 - 20x^2 + 129x - 270) \cdot 0.0500$$

$$\Rightarrow y(x) = 7 + x - 0.1667x^2 + 1.8337x - 5.0010 + 0.0500x^3$$

$$- x^2 + 6.45x - 13.5$$

$$\Rightarrow y(x) = 0.0500x^3 - 1.1667x^2 + 9.2837x - 11.5010$$

$$\therefore y(x) = 0.05x^3 - 1.1667x^2 + 9.2837x - 11.501$$

This is the required interpolating polynomial. Ans

At $x = 10$,

$$y(10) = 0.05 \times 10^3 - 1.1667 \times 10^2 + 9.2837 \times 10 - 11.501$$

$$\therefore y(10) = 14.666 \quad \underline{\underline{\text{Ans}}}$$

② Using Newton's divided difference formula, find the interpolating polynomial and hence estimate y at $x = 6$ from the given data.

$$\begin{array}{c} x : 3 \quad 7 \quad 9 \quad 10 \\ y : 160 \quad 120 \quad 72 \quad 63 \end{array}$$

sol Here,

$$\begin{array}{c} x : 3 \quad 7 \quad 9 \quad 10 \\ y : 160 \quad 120 \quad 72 \quad 63 \end{array}$$

Using Newton's divided difference formula,

$$y(x) = y_0 + (x-x_0)y[x_0, x_1] + (x-x_0)(x-x_1)y[x_0, x_1, x_2] + (x-x_0)(x-x_1)(x-x_2)y[x_0, x_1, x_2, x_3] \quad \text{--- (1)}$$

Difference table:

x	y	I DD	II DD	III DD
3	160	-10		
7	120		-2.3333	
9	72	-24	5	0.3333 1.0476
10	63	-9		

from (1),

$$y(x) = 160 + (x-3)x - 10 + (x-3)(x-7)x - 2.3333 \\ + (x-3)(x-7)(x-9)x \cdot 1.0476$$

$$\Rightarrow y(x) = 160 + (x-3)x - 10 + (x^2 - 10x + 21)x - 2.3333 + \\ (x-3)(x^2 - 16x + 63)x \cdot 1.0476$$

$$\Rightarrow y(x) = 160 - 10x + 30 - 2.3333x^2 + 23.333x - 48.9993 \\ (x^3 - 16x^2 + 63x - 3x^2 + 48x - 189)x \cdot 1.0476$$

$$\Rightarrow y(x) = -2.3333x^2 + 13.3330x + 141.0007 + 1.0476x^3 \\ - 16.7616x^2 + 65.9988x - 3.1428x^2 + 50.2848x - \\ - 197.9964$$

$$\Rightarrow y(x) = 1.0476x^3 - 19.0949x^2 + 129.6166x - 56.9957 \\ - 3.1428x^2$$

$$\therefore y(x) = 1.0476x^3 - 22.2377x^2 + 129.6166x - 56.9957$$

This is the required interpolating polynomial.

At $x=6$:

$$y(6) = 1.0476 \times 6^3 - 22.2377 \times 6^2 + 129.6166 \times 6 - \\ 56.9957$$

$$\therefore y(6) = 146.4283 \quad \underline{\underline{Ans}}$$

2. a) A curve passes through the points $(0, 18)$, $(1, 10)$, $(3, -2)$ and $(6, -12)$. Using a suitable interpolation formula, estimate y for $x = 4$.

Here,

$$\begin{array}{cccc} x: & 0 & 1 & 3 & 6 \\ y: & 18 & 10 & -2 & -12 \end{array}$$

Using Newton's divided difference formula,

$$y(x) = y_0 + (x-x_0) y[x_0, x_1] + (x-x_0)(x-x_1) y[x_0, x_1, x_2] + (x-x_0)(x-x_1)(x-x_2) y[x_0, x_1, x_2, x_3] \quad \text{--- (1)}$$

The differenced table:

x	y	I DD	II DD	III DD
0	18			
1	10	-8		
3	-2	-6	0.6667	
6	-12	-3.3333	0.5333	-0.0222

From (1),

$$y(x) = 18 + (x-0) \times -8 + (x-0)(x-1) \times 0.6667 + (x-0)(x-1)(x-3) \times -0.0222$$

$$\Rightarrow y(x) = 18 - 8x + (x^2 - x) \times 0.6667 + x(x^2 - 4x + 3) \times -0.0222$$

$$\Rightarrow y(x) = 18 - 8x + 0.6667x^2 - 0.6667x + (x^3 - 4x^2 + 3x) \times -0.0222$$

$$\Rightarrow y(x) = -0.0222x^3 + 0.7555x^2 - 8.7332x + 18$$

$$\Rightarrow y(x) = -0.0222x^3 + 0.7555x^2 - 8.7332x + 18$$

This is the required interpolating polynomial. Ans

At $x = 4$;

$$y(4) = -0.0222 \times 4^3 + 0.7555 \times 4^2 - 8.7332 \times 4 + 18$$

$$\therefore y(4) = -6.2656 \quad \underline{\underline{\text{Ans}}}$$

② Newton's Interpolation Formula: (Forward/Backward)

③ Using Newton's interpolation formula, estimate y at $x=1.15$ and $x=2$ from the given table.

$$x : 1 \quad 1.4 \quad 1.8 \quad 2.2$$

$$y : 3.5 \quad 4.82 \quad 5.96 \quad 6.5$$

Here, ...

$$x : 1 \quad 1.4 \quad 1.8 \quad 2.2$$

$$y : 3.5 \quad 4.82 \quad 5.96 \quad 6.5$$

The difference Table is:

x	y	$1D$	$2D$	$3D$
1	3.5			
1.4	4.82	1.32		
1.8	5.96	1.14	-0.18	
2.2	6.5	0.54	-0.6	-0.42

To estimate y at $x=1.15$, we apply Newton's Forward interpolation formula,

$$y(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots \quad (1)$$

where,

$$p = \frac{x - x_0}{h} = \frac{1.15 - 1}{0.4} = 0.3750$$

$$\therefore y(1.15) = 3.5 + 0.375 \times 1.32 + \frac{0.375(0.375-1)}{2!} \times -0.18 + \frac{0.375(0.375-1)(0.375-2)}{3!} \times -0.42$$

$$\therefore y(1.15) = 3.9894. \quad \underline{\underline{\text{Ans}}}$$

To estimate y at $x=2$, we apply Newton's Backward interpolation formula,

$$y(x) = y_n + q \nabla y_n + \frac{q(q+1)}{2!} \nabla^2 y_n + \frac{q(q+1)(q+2)}{3!} \nabla^3 y_n + \frac{q(q+1)(q+2)(q+3)}{4!} \nabla^4 y_n + \dots \quad (2)$$

where,

$$q = \frac{x - x_n}{h} = \frac{2 - 2.2}{0.4} = -0.5$$

$$\therefore y(2) = 6.5 + (-0.5) \times 0.54 + \frac{-0.5(-0.5+1)}{2!} \times -0.6 + \frac{-0.5(-0.5+1)(-0.5+2)}{3!} \times -0.42$$

$$\therefore y(2) = 6.3313 \quad \underline{\underline{\text{Ans}}}$$

② From the given data, estimate the number of students whose marks lies between 40-45 and 65-80.

Marks :	30-40	40-50	50-60	60-70	70-80
No. of students:	31	42	51	35	31

Here,

Marks :	30-40	40-50	50-60	60-70	70-80
No. of students :	31	42	51	35	31

changing the given data into c.f. and calculating difference:

less than marks (x)	C.f. (y)	ID	IID	IIID	IIIID
40	31	42	9	-25	37
50	73	51	-16	12	
60	124	35	-4		
70	159	31			
80	190				

To estimate y at $x = 45$, we apply Newton's Forward Interpolation Formula,

$$y(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots \quad (1)$$

where, $p = \frac{x - x_0}{h} = \frac{45 - 40}{10} = 0.5$

From (1),

$$\begin{aligned} \Rightarrow y(45) &= 31 + 0.5 \times 42 + \frac{0.5(0.5-1)}{2!} \times 9 + \frac{0.5(0.5-1)(0.5-2)}{3!} \times -25 \\ &\quad + \frac{0.5(0.5-1)(0.5-2)(0.5-3)}{4!} \times 37 \\ &= \frac{789}{16} - \frac{185}{128} \end{aligned}$$

$$\therefore y(45) = 47.8672 \approx 48$$

∴ The no. of students whose marks is less than 45 is ≈ 48 and 40 is ≈ 31 .

∴ The no. of students whose marks lies between 40-45 is $\approx (48-31) \approx 17$. Ans

Again, To find the y at $x = 65$, we apply Newton's Backward interpolation formula,

$$y(x) = y_n + q \nabla y_n + \frac{q(q+1)}{2!} \nabla^2 y_n + \frac{q(q+1)(q+2)}{3!} \nabla^3 y_n + \frac{q(q+1)(q+2)(q+3)}{4!} \nabla^4 y_n + \dots \quad (2)$$

where,

$$q = \frac{x - x_n}{h} = \frac{65 - 80}{10} = -1.5$$

From (2),

$$y(65) = 190 + (-1.5) \times 31 + \frac{-1.5(-1.5+1)}{2!} \times -4 + \frac{-1.5(-1.5+1)(-1.5+2)}{3!} \times 37 + \frac{-1.5(-1.5+1)(-1.5+2)(-1.5+3)}{4!} \times 37$$

$$\Rightarrow y(65) = \frac{571}{4} + \frac{111}{128}$$

$$\therefore y(65) = 143.6172 \approx 144$$

∴ The no. of students whose marks is less than 65 ≈ 144 and 80 ≈ 190 .

∴ The no. of students whose marks lies between (65-80) is $\approx (190-144) \approx 46$. Ans

2023-Spring

2 a) From the data given below, find the number of students whose weight is between 60 to 70.

weight in lbs: 0-40 40-60 60-80 80-100 100-120

No. of students: 250 120 100 70 50.

Here,

weight in lbs : 0-40 40-60 60-80 80-100 100-120

No. of students : 250 120 100 70 50

Now, changing the above data into c.f. table, and calculating differences:

less than weight (x)	No. of student (y)	I D	II D	III D	IV D	V D
20	125	125				
40	250		-5			
60	370	120	-20	-15	5	
80	470	100	-30	-10	20	15
100	540	70	-20	10		
120	590	50				

To estimate y at $x = 70$, we apply, Newton's Backward Interpolation Formula,

$$y(x) = y_n + q \nabla y_n + \frac{q(q+1)}{2!} \nabla^2 y_n + \frac{q(q+1)(q+2)}{3!} \nabla^3 y_n + \frac{q(q+1)(q+2)(q+3)}{4!} \nabla^4 y_n + \dots \quad (1)$$

where, $q = \frac{x - x_n}{h} = \frac{70 - 120}{20} = -2.5$

from (1),

$$y(70) = 590 + (-2.5) \times 50 + \frac{-2.5(-2.5+1)}{2!} \times (-20) + \frac{-2.5(-2.5+1)(-2.5+2)}{3!} \times 10 + \frac{-2.5(-2.5+1)(-2.5+2)(-2.5+3)}{4!} \times 20 + \frac{-2.5(-2.5+1)(-2.5+2)(-2.5+3)(-2.5+4)}{5!} \times 15$$

$$= \frac{3395}{8} - \frac{25}{32} + \frac{45}{256}$$

$$\therefore y(70) = 423.418 \approx 423$$

~~Hence~~

$\therefore$ The no. of students whose marks is less than 70 ≈ 423 and 80 is ≈ 470 .

Hence, the no. of students whose marks lies between (70-80) is $\approx (470-423) \approx 47$. ~~Ans~~

2016-Fall

2.6) The following table gives the population of a town during the last six censuses. Estimate the increase in the population during the period from 1976 to 1978.

Year: 1941 1951 1961 1971 1981 1991
 Popⁿ : 12 15 20 27 39 52

Here,

Year (x): 1941 1951 1961 1971 1981 1991

Popⁿ (y): 12 15 20 27 39 52

The difference table:

Year(x)	Pop ⁿ (y)	API ID	IID	IIID	IVD	V D
1941	12	3				
1951	15		2			
1961	20	5		0		
1971	27	7	2	3		
1981	39	12	5	-4	-7	
1991	52	13	1			-10

To estimate y at x = 1976, we apply Newton's Backward Interpolation formula,

$$y(x) = y_n + q \nabla y_n + \frac{q(q+1)}{2!} \nabla^2 y_n + \frac{q(q+1)(q+2)}{3!} \nabla^3 y_n + \frac{q(q+1)(q+2)(q+3)}{4!} \nabla^4 y_n + \frac{q(q+1)(q+2)(q+3)(q+4)}{5!} \nabla^5 y_n$$

where, $q = \frac{x - x_n}{h} = \frac{1976 - 1991}{10} = -1.5$

$$\begin{aligned} y(1976) &= 52 + (-1.5) \times 13 + \frac{(-1.5)(-1.5+1)}{2!} \times 1 + \frac{(-1.5)(-1.5+1)(-1.5+2)}{3!} \times (-7) \\ &\quad + \frac{(-1.5)(-1.5+1)(-1.5+2)(-1.5+3)}{4!} \times (-4) + \frac{(-1.5)(-1.5+1)(-1.5+2)(-1.5+3)(-1.5+4)}{5!} \times (-10) \end{aligned}$$

$$= \frac{261}{8} - \frac{21}{128} - \frac{15}{128}$$

$$\therefore y(1976) = 32.34 \approx 32$$

To estimate y at $x = 1978$, we apply Newton's Backward Interpolation Formula,

$$y(x) = y_n + q \nabla y_n + \frac{q(q+1)}{2!} \nabla^2 y_n + \frac{q(q+1)(q+2)}{3!} \nabla^3 y_n \\ + \frac{q(q+1)(q+2)(q+3)}{4!} \nabla^4 y_n + \frac{q(q+1)(q+2)(q+3)(q+4)}{5!} \nabla^5 y_n$$

$$\text{where, } q = \frac{x - x_n}{h} = \frac{1978 - 1991}{10} = -1.3$$

$$y(1978) = 52 + (-1.3) \times 13 + \frac{-1.3(-1.3+1)}{2!} x^1 + \frac{-1.3(-1.3+1)(-1.3+2)}{3!} x^{-4} \\ + \frac{-1.3(-1.3+1)(-1.3+2)(-1.3+3)}{4!} x^{-7} + \frac{-1.3(-1.3+1)(-1.3+2)(-1.3+3)(-1.3+4)}{5!} x^{-10}$$

$$\Rightarrow y(1978) = \cancel{35.1130} - 0.1354 - 0.1044$$

$$\therefore y(1978) = 34.8833 \approx 35$$

Hence, the no. of

$\therefore$ The ~~no. of~~ population at 1976 is ≈ 32 and population at 1978 is ≈ 35 .

$\therefore$ The increase in the population during the period ~~de~~ from 1976 to 1978 is $\approx (35 - 32)$
 $\approx \underline{\underline{3 \text{ Ans}}}$

Curve Fitting
2021-Spring

① Given that:

$x: 1 \quad 2 \quad 3 \quad 4 \quad 5$

$y: 14 \quad 27 \quad 401 \quad 55 \quad 68$

Find the equation of straight line using least square method.

Sol Here,

$x: 1 \quad 2 \quad 3 \quad 4 \quad 5$

$y: 14 \quad 27 \quad 401 \quad 55 \quad 68$

Let $y = ax + b$ ——— (1)

be the equation of straight linear equation.

The values of a and b are obtained by solving these equations,

$$\sum y = na + b \sum x \text{ ——— (2)}$$

$$\sum xy = a \sum x + b \sum x^2 \text{ ——— (3)}$$

Now,

$$n = 5$$

x	y	x^2	xy
1	14	1	14
2	27	4	54
3	401	9	1203
4	55	16	220
5	68	25	340
15	565	55	1831

From (2) and (3), we get:-

$$565 = 5a + 15a$$

$$1831 = 15a + 55b$$

On solving these equation, we get;

$$a = 72.2$$

$$b = 13.6$$

From (1),

$$y = 72.2 + 13.6x$$

This is the required equation of linear equation. Ans

Q Given that:

$$x: 1 \quad 2 \quad 3 \quad 4 \quad 5$$

$$y: 2 \quad 6 \quad 7 \quad 8 \quad 10$$

Fit a second degree polynomial of the form $y = a + bx + cx^2$ to the given data.

sol Here,

$$x: 1 \quad 2 \quad 3 \quad 4 \quad 5$$

$$y: 2 \quad 6 \quad 7 \quad 8 \quad 10$$

we have,

$$y = a + bx + cx^2 \quad \text{--- (1)}$$

The values of a , b and c are obtained by solving these equations, we get:

$$\Sigma y = na + b\Sigma x + c\Sigma x^2 \quad \text{--- (2)}$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3 \quad \text{--- (3)}$$

$$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4 \quad \text{--- (4)}$$

Now,
 $n = 5$

x	y	xy	x^2	x^2y	x^3	x^4
1	2	2	1	2	1	1
2	6	12	4	24	8	16
3	7	21	9	63	27	81
4	8	32	16	128 128	64	256
5	10	50	25	250	125	625
15	33	117	55	467	225	979

from (2), (3) and (4), we get:

$$33 = 5a + 15b + 55c$$

$$117 = 15a + 55b + 225c$$

$$467 = 55a + 225b + 979c$$

on solving these equations, we get:

$$a = -0.8$$

$$b = 3.5143$$

$$c = -0.2857$$

from (1),

$$y = -0.8 + 3.5143x - 0.2857x^2$$

This is the required second degree polynomial. Ans

1. Fit an exponential equation of the form $y = ae^{bx}$ using least square method from the following table.

x:	1	2	3	4	5	6
y:	1.65	2.7	4.5	7.35	12.2	15

Here,

x:	1	2	3	4	5	6
y:	1.65	2.7	4.5	7.35	12.2	15

We have,

$$y = ae^{bx} \quad \text{--- (1)}$$

Taking $\ln$ on both sides, we get:

$$\ln y = \ln(ae^{bx})$$

$$\Rightarrow \ln y = \ln a + bx \ln e$$

$$\Rightarrow \ln y = \ln a + bx \quad (\because \ln e = 1)$$

$$\text{Put } Y = A + BX \quad \text{--- (2)}$$

where,

$$Y = \ln y$$

$$A = \ln a$$

$$B = b$$

$$X = x$$

The values of A and B are obtained by solving these equations,

$$\sum Y = nA + B \sum X \quad \text{--- (3)}$$

$$\sum XY = A \sum X + B \sum X^2 \quad \text{--- (4)}$$

where $n = 6$

$X=x$	y	$Y=\ln y$	XY	x^2
1	1.65	0.5008	0.5008	1
2	2.7	0.9933	1.9865	4
3	4.5	1.5041	4.5122	9
4	7.35	1.9947	7.9788	16
5	12.2	2.5014	12.5072	25
6	15	2.7081	16.2483	36
$\Sigma X = 21$		10.2023	43.7338	91

from (2) & (3), we get;

$$10.2023 = 6A + 21B$$

$$43.7338 = 21A + 91B$$

On solving these equations, we get;

$$A = 0.0952$$

$$B = 0.4586$$

We have,

$$A = \ln a \Rightarrow a = e^A = e^{0.0952} = 1.0999 \approx 1.1$$

$$B = b \therefore b = B = 0.4586$$

from (1),

$$y = 1.1e^{0.4586x}$$

This is the required exponential equation. Ans

2013-Fall

②. Fit a ~~curve~~ power function $y = ax^b$ to the data.

x : 2 4 6 8
 y : 1.4 2 2.4 2.6

Sol Here,

x : 2 4 6 8
 y : 1.4 2 2.4 2.6

we have,

$$y = ax^b \text{ ————— (1)}$$

Taking $\ln$ on both sides, we get,

$$\ln y = \ln(ax^b)$$

$$\Rightarrow \ln y = \ln a + b \ln x$$

$$\text{put } Y = A + BX \text{ ————— (2)}$$

where,

$$Y = \ln y$$

$$A = \ln a$$

$$B = b$$

$$X = \ln x$$

The values of A and B are obtained by solving these equations.

$$\Sigma Y = nA + B \Sigma X \text{ ————— (3)}$$

$$\Sigma XY = A \Sigma X + B \Sigma X^2 \text{ ————— (4)}$$

where,

$$n = 4$$

x	x y	$X = \ln x$	$Y = \ln y$	XY	x^2
2	1.4	0.6931	0.3365	0.2332	0.4805
4	2	1.3863	0.6931	0.9608	1.9218
6	2.4	1.7918	0.8755	1.5687	3.2104
8	2.6	2.0794	0.9555	1.9869	4.3241
		5.9506	2.8606	2.7497	9.9367

From (3) and (4), we get;

$$2.8606 = 4A + 5.9506B$$

$$4.7497 = 5.9506A + 9.9367B$$

on solving these equations, we get;

$$A = 0.0372$$

$$B = 0.4557 = b$$

$$\therefore A = \ln a \Rightarrow a = e^A = e^{0.0372} \approx 1.0379$$

From (1),

$$y = 1.0379 \cdot x^{0.4557}$$

This is the required power function. Ans

2019 - Fall

③ Fit a curve of the form $PV^\alpha = B$ by using least square method, where P = pressure; V = volume.

$$P: 10 \quad 20 \quad 50 \quad 100 \quad 150$$

$$V: 38.4 \quad 20 \quad 8.51 \quad 4.44 \quad 3.03$$

sol Here,

$$P: 10 \quad 20 \quad 50 \quad 100 \quad 150$$

$$V: 38.4 \quad 20 \quad 8.51 \quad 4.44 \quad 3.03$$

We have,

$$PV^\alpha = B \quad \text{--- (1)}$$

$$\Rightarrow \ln(PV^\alpha) = \ln B \quad (\because \text{Taking } \ln \text{ on both sides})$$

$$\Rightarrow \ln P + \alpha \ln V = \ln B$$

$$\Rightarrow \alpha \ln V = \ln B - \ln P$$

$$\Rightarrow \therefore \ln V = \frac{1}{\alpha} \ln B - \frac{1}{\alpha} \ln P$$

$$\text{Let } Y_A = A + BX \quad \text{--- (2)}$$

where,

$$A Y = \ln v$$

$$A = \frac{1}{\alpha} \ln p$$

$$B = -\frac{1}{\alpha}$$

$$x = \ln p$$

The values of A and B are obtained by solving these equations,

$$\sum Y = nA + B \sum x \quad \text{--- (3)}$$

$$\sum xY = A \sum x + B \sum x^2 \quad \text{--- (4)}$$

where, $n = 5$

P	v	$x = \ln p$	$Y = \ln v$	xY	x^2
10	38.4	2.3026	3.6481	8.4000	5.3019
20	20	2.9957	2.9957	8.9744	8.9744
50	8.51	3.9120	2.1412	8.3766	15.3039
100	4.44	4.6052	1.4907	6.8647	21.2076
150	3.03	5.0106	1.1086	5.5546	5.0106
		18.8261	11.3842	38.1703	75.8943

From (3) and (4), we get:

$$11.3842 = 5A + 18.8261B$$

$$38.1703 = 18.8261A + 75.8943B$$

On solving these equations, we get:-

$$A = 5.8049 ; B = -0.9370$$

We have,

$$B = -\frac{1}{\alpha} \Rightarrow \alpha = -\frac{1}{B} = \frac{1}{-0.9370} = 1.0672$$

$$A = \frac{1}{\alpha} \ln p$$

$$\Rightarrow \ln p = A\alpha \Rightarrow p = e^{A\alpha}$$

$$\Rightarrow p = e^{5.8049 \times 1.0672}$$

$$\therefore p = 490.2862$$

from (1);

$$PV^{1.0672} = 490.2862$$

This is the required curve. Ans

2019-Spring

④ The time and voltage are given by

$$T(t): \quad 0 \quad 2 \quad 4 \quad 6 \quad 8$$

$$V: \quad 150 \quad 63 \quad 28 \quad 12 \quad 5.6$$

Fit a curve of the form $V = ae^{bt}$ using least square method.

sol Here,

$$T(t): \quad 0 \quad 2 \quad 4 \quad 6 \quad 8$$

$$V: \quad 150 \quad 63 \quad 28 \quad 12 \quad 5.6$$

we have,

$$V = ae^{bt} \quad \text{--- (1)}$$

Taking $\ln$ on both sides, we get:-

$$\ln V = \ln(ae^{bt})$$

$$\Rightarrow \ln V = \ln a + bt \ln e$$

$$\Rightarrow \ln V = \ln a + bt \quad (\because \ln e = 1)$$

$$\text{put } Y = A + BX \quad \text{--- (2)}$$

where,

$$Y = \ln V$$

$$A = \ln a$$

$$B = b$$

$$X = t$$

The values of ~~an~~ A and B are obtained by solving these equations,

$$\Sigma Y = nA + B \Sigma X \quad \text{--- (3)}$$

$$\Sigma XY = A \Sigma X + B \Sigma X^2 \quad \text{--- (4)}$$

where, $n = 5$

$X=t$	V	$Y=\ln V$	XY	x^2
0	150	5.0106	0	0
2	63	4.1431	8.2863	4
4	28	3.3322	13.3288	16
6	12	2.4849	14.9094	36
8	5.6	1.7228	13.7821	64
20		16.6936	50.3067	120

From (3) and (4), we get:-

$$16.6936 = 5A + 20B$$

$$50.3067 = 20A + 120B$$

On solving these equations, we get:-

$$A = 4.9855$$

$$B = -0.4117 = b$$

$$\therefore A = \ln a \Rightarrow a = e^A = e^{\cancel{0.4117} 4.9855}$$

$$\therefore a = \cancel{0.6625} 146.2767$$

From (1),

$$\cancel{y = 0.6625e}$$

$$\therefore y = 146.2767 e^{-0.4117t}$$

This is the required curve. Ans